

MOMENTUM ARENA

Complete Feature Guide

Version 2.0 | July 2026

Mathura's Premier Multi-Sport Arena

Web · iOS App · Android App — every feature with its flow

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1. PLATFORM OVERVIEW

Momentum Arena runs one product on three surfaces backed by a single system:

- **Web** — www.momentumarena.com (customer site + full admin console).
- **iOS + Android apps** — customer experience plus the complete admin console (full parity with web admin), distributed via TestFlight / App Store and Google Play.
- **Environments** — production (main branch) and staging (development branch at development.momentumarena.com); every git push deploys automatically with schema-atomic builds.

Business modules

Court booking (cricket, football, pickleball + bowling machine), cafe with live kitchen board, pickup shop with POS, Momentum Points rewards, coupons engine, waitlist, recurring bookings, push notifications, expenses tracking, and a full analytics stack (first-party events + Google Analytics 4 on all three surfaces).

Tech stack

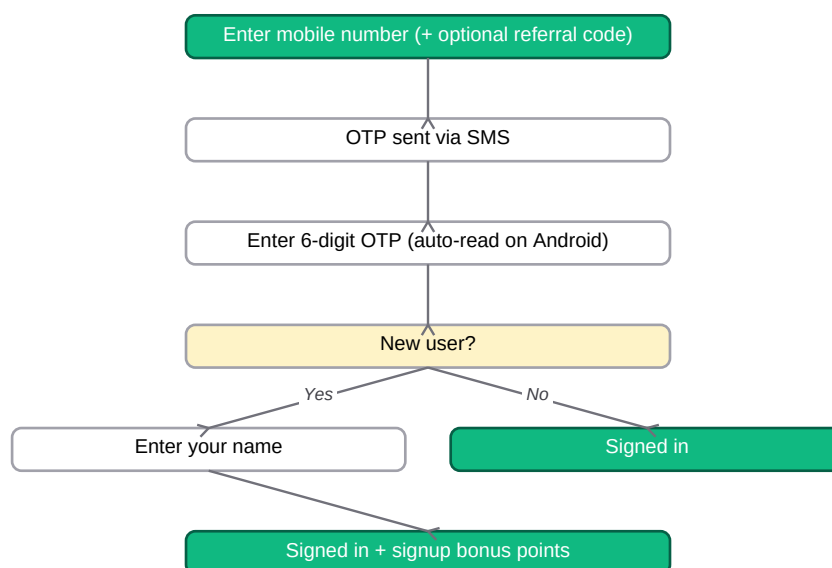
Next.js App Router + TypeScript + Prisma/PostgreSQL (Neon) on Vercel · React Native apps (self-hosted OTA updates) · Firebase (push + analytics) · Razorpay and PhonePe payments · MSG91 (SMS OTP + email).

2. CUSTOMER — AUTHENTICATION & ONBOARDING

One auth model on all surfaces: mobile-number OTP. No passwords for customers.

- Login modal (web) / login screens (app) — 10-digit phone, 6-digit OTP via MSG91 SMS.
- Android auto-reads the OTP (SMS User Consent); resend with cooldown.
- New users are asked for their name right after OTP verification, then land signed-in.
- Optional referral code at signup — both sides earn bonus Momentum Points.
- Server-side abuse protection: per-phone rate limits + failed-attempt lockouts.
- Browsing is open — login is required only at the payment step (checkout) and for account pages.
- In-app account deletion (store-compliance) with full data cleanup.

FLOW: Sign in / sign up



3. CUSTOMER — HOME & NAVIGATION

Web: / · App: Home tab

- Hero with sport tiles (Cricket, Football, Pickleball) and Order Food CTA.
- Signed-in home shows upcoming bookings and the rewards points chip.
- Pickleball launch-promo banner appears automatically while its coupon is live.
- Facilities, contact, Google Maps, WhatsApp + social links (web).
- App bottom tabs: Home · Sports · Cafe · Shop · Account. Web mirrors with header/bottom nav.
- Force-update gate in the app: admin can require a minimum version; soft-update nags once per session.

4. CUSTOMER — COURT BOOKING

Web: /book → /book/[sport] → slots → /book/checkout · App: Sports tab

Choose sport & court

- Sports: Cricket, Football, Pickleball (admin can add court configs per sport).
- Court options with size diagrams (e.g. half-field 'venue assigns a side'); single-option sports skip straight to slots.
- Cricket additionally offers the Bowling Machine (separate flow, section 5).

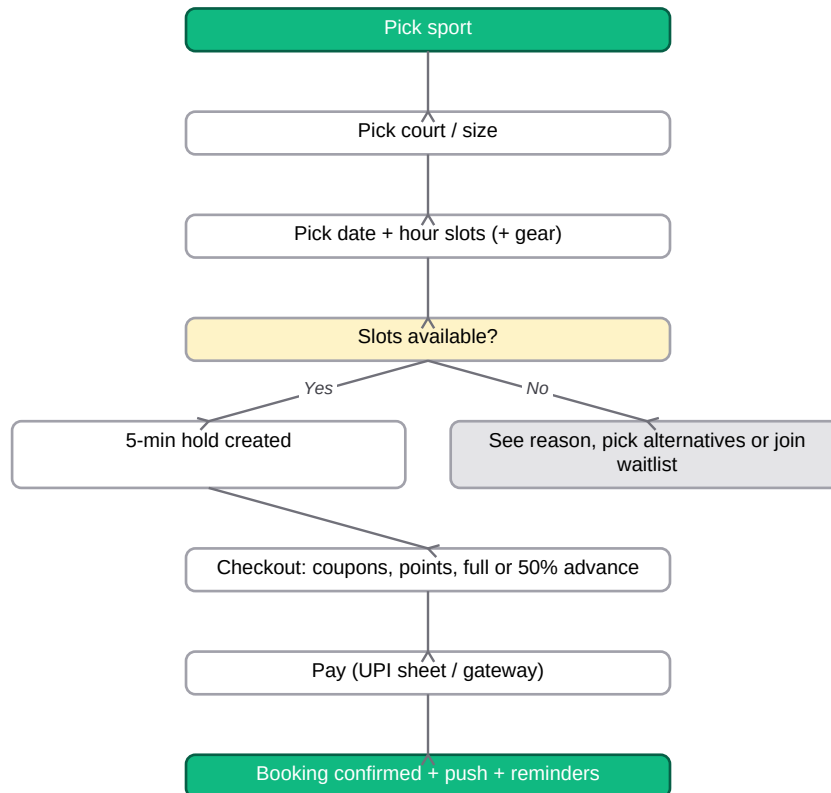
Pick date & slots

- 30-day date strip; hourly grid 5 AM – 1 AM with per-slot prices (peak / off-peak / weekend via admin time classifications).
- Any combination of available hours — multi-select, no contiguity restriction.
- Full slots are tappable: see WHY it's blocked plus alternative courts/times, or join the waitlist (section 6).
- Late-night 12–1 AM slot appears on the night it belongs to (display-shifted).
- Rental gear picker (e.g. bats, machine balls) — per-slot pricing, snapshotted onto the booking.

Hold & lock

- Selecting slots creates a 5-minute hold with a live countdown at checkout — no one can double-book underneath you.
- Conflicts caught at lock time list exactly which hours were just taken.
- Expired hold = friendly 'slot released' alert and a clean restart.

FLOW: Book a court



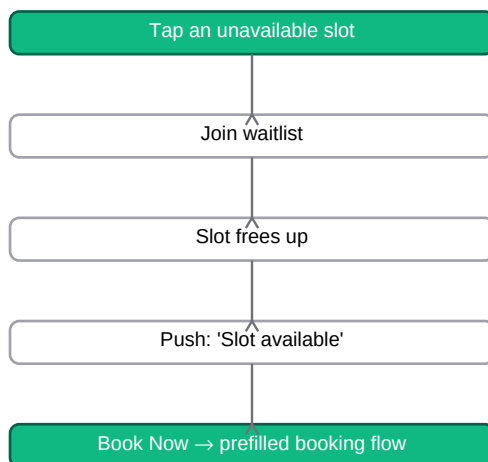
5. CUSTOMER — BOWLING MACHINE BOOKING

- 30-minute slots (vs hourly courts) with contiguous-selection enforcement.
- Own availability grid + pricing; machine-ball rental add-on.
- Same hold → checkout → payment pipeline as courts.

6. CUSTOMER — WAITLIST

- Tap a full future slot → join the waitlist for that court + hour.
- The moment the slot frees (cancellation/admin change), everyone whose range covers it gets a push notification.
- Waitlist screen under Account lists entries with one-tap Book Now (prefilled date/slot).
- Entries can be cancelled anytime; joins are tracked in analytics for demand insight.

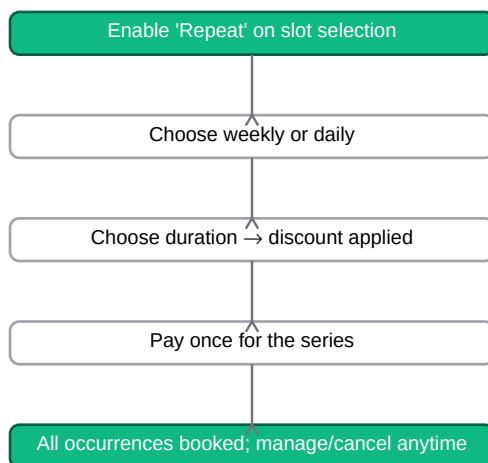
FLOW: Waitlist



7. CUSTOMER — RECURRING BOOKINGS

- Weekly or daily series with admin-configured duration discounts (e.g. 4 weeks = X% off).
- Series pricing shown up front; one payment covers the series.
- Manage & cancel the series from the app (Recurring screen).
- Admin has a matching recurring console (create/manage on behalf of customers).

FLOW: Recurring series



8. CUSTOMER — CHECKOUT, DISCOUNTS & PAYMENTS

Checkout

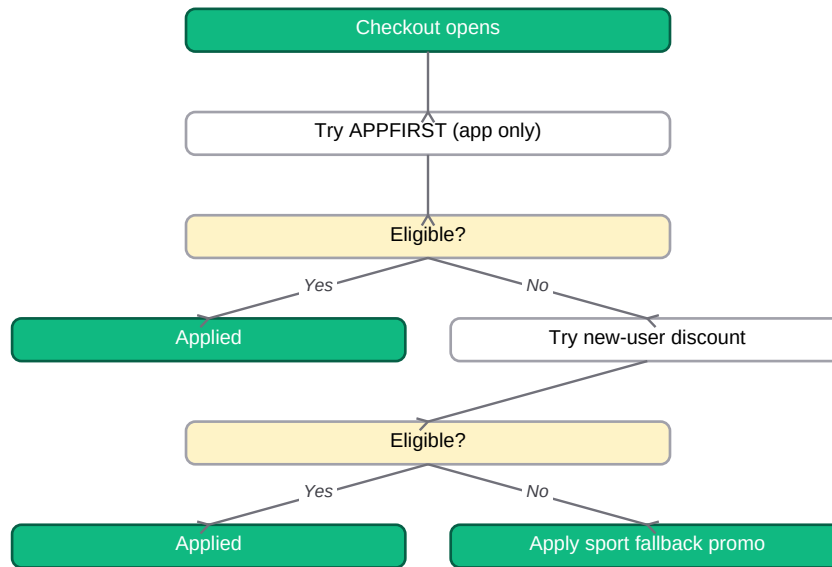
- Booking summary with per-slot breakdown, gear, points discount and live total.
- Amount choice: Pay Full or Pay 50% Advance (remainder at venue) — admin can toggle advance off.
- Earn-preview: 'You'll earn N Momentum Points' updates live with the payable amount.

Discounts (auto-apply chain)

- APPFIRST — first-ever app booking discount (app-only), attempted automatically & safely.
- New-user discount — auto-applied when eligible.

- Sport fallback promos (e.g. PICKLEBALL25, FLAT100) — auto-applied when nothing better fits.
- Manual code entry + a browsable coupon drawer (targeted coupons via user groups).

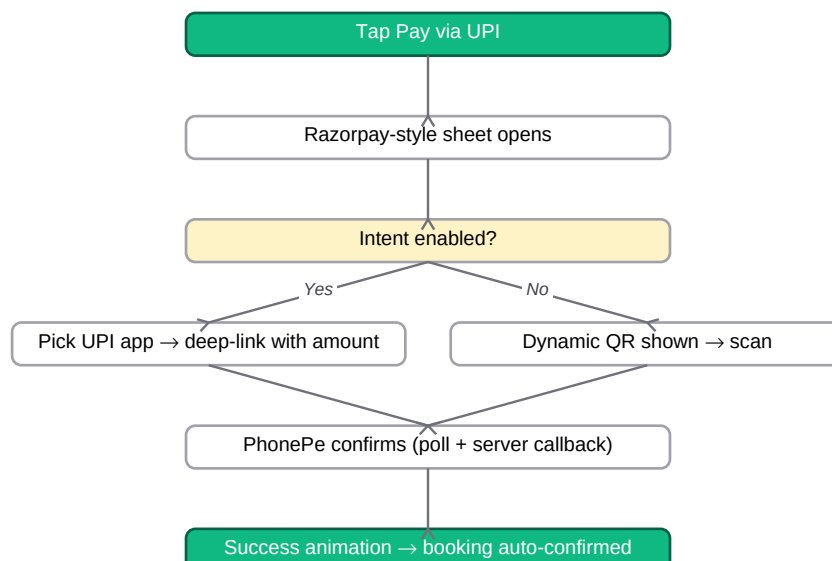
FLOW: Auto-apply discount chain



Payment methods

- **UPI (recommended, zero gateway fee)** — a Razorpay-style dark bottom sheet with three admin-selectable modes:
- **UPI Intent**: pick PhonePe / GPay / Paytm / BHIM / Other → app opens with amount prefilled → auto-confirm via PhonePe Dynamic QR + status polling.
- **Scan QR**: dynamic QR bound to this exact payment; auto-confirms, no screenshots.
- **Static QR fallback**: pay to venue VPA → booking pends → WhatsApp screenshot → admin UTR verification.
- **Card / Netbanking** — Razorpay native sheet (web + app), signature-verified server-side.
- Admin controls which of these are live from Payment Settings (section 20).
- Safety nets: server-to-server payment callbacks, webhook capture of orphaned payments, payment-recovery queue.

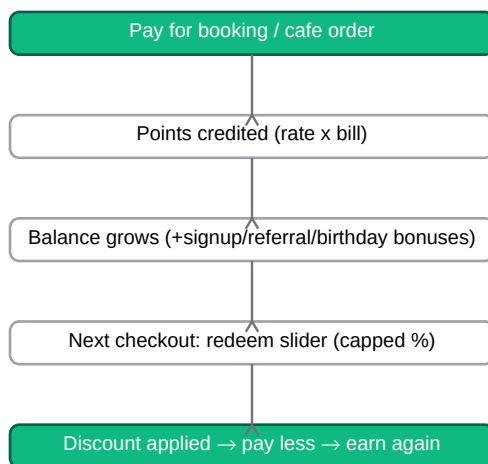
FLOW: UPI payment (dynamic QR)



9. CUSTOMER — MOMENTUM POINTS

- Earn on bookings and cafe orders (admin-set rates per sport / cafe), plus signup, referral and birthday bonuses.
- Advance-pay bookings earn on the advance now and the remainder when the venue collects it.
- Redeem at checkout with a live slider — capped at an admin-set % of the bill, minimum redeem threshold.
- Rewards screen: balance, full transaction history, 'how it works' explainer; points chip on home/account.
- Every earn triggers a push (each type individually editable/toggleable by admin — section 14).

FLOW: Earn & redeem loop



10. CUSTOMER — MY BOOKINGS & ACCOUNT

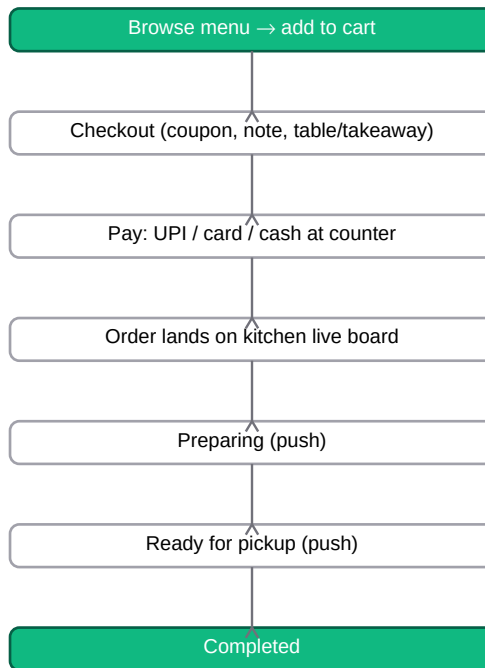
- Bookings list: upcoming vs past, infinite scroll, status badges (confirmed / pending / completed / absent / cancelled / refunded).
- Booking detail: full breakdown, payment info, add-to-calendar (ICS) export; invoice download on web.
- Account hub: edit name, Momentum Points tile, My Coupons browser, Waitlist, policies, sign-out, delete account.
- Confirmation push on booking + reminders at 24h / 2h / 1h before play.

11. CUSTOMER — CAFE ORDERING

Web: /cafe · App: Cafe tab

- Live menu grouped by category with veg/non-veg marks, photos and real-time stock (sold-out greys out; stock caps quantity).
- Admin can close the cafe → warm 'closed' screen with hours.
- Cart with quantity steppers → checkout with coupon field and order note.
- Guests can order without an account (name + phone optional on web).
- Pay via UPI (same dynamic-QR sheet as bookings), card/netbanking, or cash at counter.
- Order tracking screen polls kitchen status; push when the kitchen starts preparing and when it's ready.
- Earn Momentum Points on cafe spend; cafe-scoped coupons supported.

FLOW: Cafe order



12. CUSTOMER — SHOP

- Pickup-at-venue product catalog (equipment, merch) on web + app.
- Product detail with images and stock; order online, collect at the venue.
- Same payment stack as cafe; order history with status tracking.
- Admin-side POS covers walk-in sales (section 18).

13. CUSTOMER — CHAT ASSISTANT

- Built-in assistant (web widget + full app tab) answering from the FAQ knowledge base.
- Intent matching with quick-action buttons that deep-link into booking, cafe, rewards etc.
- FAQ content is fully admin-managed (categories, keywords, ordering).

14. PUSH NOTIFICATIONS & AUTO-PUSH TEMPLATES

Firestore Cloud Messaging to both apps, with tap-routing to the right screen. Two halves:

Automated pushes (20 registered triggers)

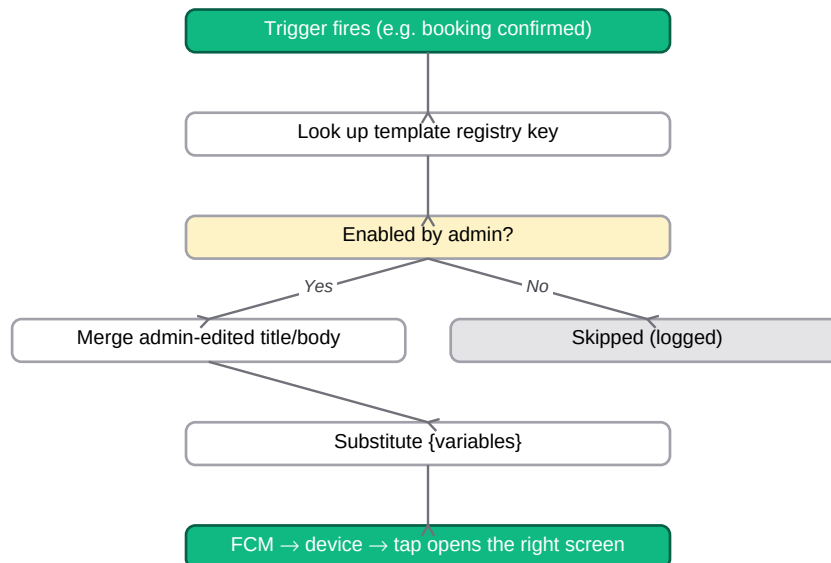
- Customer: booking confirmed / cancelled / refunded, 24h / 2h / 1h reminders, waitlist slot available, cafe preparing / ready, 7 rewards-earned variants.
- Admin devices: new pending booking, booking confirmed / cancelled / refunded alerts.

Admin-configurable templates

- Every automated push is editable from Admin → Auto Push Messages (web + app): title, body, per-push enable/disable.
- Documented {variables} per message with insert-chips and live preview; typo-guard rejects unknown placeholders.

- Edits apply on the very next send — no deploy. New triggers MUST register in the template registry, so future pushes are automatically configurable.
- Manual push console: broadcast or targeted sends (user groups / individual), plus delivery analytics.

FLOW: Automated push pipeline



15. ADMIN — ACCESS, ROLES & PERMISSIONS

Web: [/godmode](#) → [/admin](#) · App: [hidden 5-tap entry on the Account screen](#)

- Separate admin session (independent of the customer session).
- Roles: SUPERADMIN (everything, undeletable), ADMIN (granular permissions), STAFF (restricted to day-to-day booking ops + live cafe board).
- Admin invites via email with expiring set-password links; password reset + superadmin-alert emails.
- The mobile admin console mirrors web admin at full parity, permission-gated per screen and per API route.

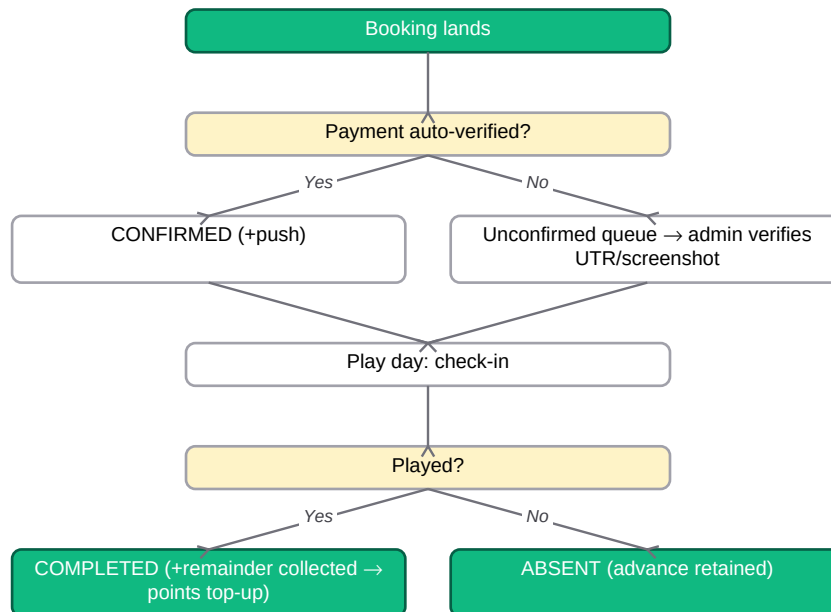
Permissions: MANAGE_BOOKINGS · MANAGE_PRICING · MANAGE_SLOTS · MANAGE_SPORTS · MANAGE_USERS · MANAGE_DISCOUNTS · MANAGE_FAQS · VIEW_ANALYTICS · VIEW_RAZORPAY · MANAGE_ADMIN_USERS (superadmin-only) · MANAGE_CAFE_MENU · MANAGE_CAFE_ORDERS · MANAGE_CAFE_DISCOUNTS · MANAGE_REWARDS · MANAGE_COUPONS · MANAGE_EXPENSES · MANAGE_PUSH · MANAGE_SHOP_CATALOG · MANAGE_SHOP_ORDERS · MANAGE_APP_RELEASES · MANAGE_PAYMENT_SETTINGS

16. ADMIN — BOOKINGS OPERATIONS

- **All Bookings:** search, multi-select status/sport filters, sort options, pagination; detail page with full history log.
- **Unconfirmed queue:** pending payments (static-QR/UTR & advance flows) for one-tap verify/confirm.
- **Calendar:** full 24-hour day grid across all courts, color-coded, tap-to-inspect, off-hours admin bookings priced at PEAK.
- **Create booking:** book on behalf of walk-in/phone customers, any hour of the day, any payment method incl. cash/free.
- **Check-in:** mark arrivals; complete or mark ABSENT after the slot (absent shows advance-only owed).
- **Extend +30 min:** stretch a live booking when the next slot is free, price auto-added and logged.
- **Cancel / refund** with reason (customer gets the push); every mutation logged with who/when.

- **UTR verification** screen for static-QR payments; **Payment Recovery** for money-captured-but-no-booking cases (section 20).
- Recurring console for series management.

FLOW: Booking lifecycle (ops view)



17. ADMIN — COURTS, PRICING & SLOT INVENTORY

- **Sports & courts:** court configurations per sport (sizes, zones, half-court behaviour), enable/disable instantly.
- **Equipment:** rentable gear catalog with per-slot pricing, attached to sports.
- **Pricing matrix:** per court x time-classification (weekday off-peak / peak / weekend...), inline editing.
- **Time classifications:** admin-defined day/hour bands that drive slot pricing.
- **Slot blocks:** block hours or whole days per court / sport / venue for maintenance and events.
- **Generator:** bulk slot/pricing generation utility.

18. ADMIN — CAFE & SHOP OPERATIONS

Cafe

- Menu CRUD with images, veg/non-veg, tags, category ordering and stock counts (auto-decrement on order).
- Cafe open/closed master switch.
- Orders list + stats; **Live kitchen board:** Pending → Preparing → Ready kanban, auto-refresh, sound alert, fullscreen KDS mode.
- Create order (walk-in), cafe-scoped coupons.

Shop

- Product catalog CRUD with stock, product orders management,
- **Walk-in POS:** ring up in-person sales with any payment method.

19. ADMIN — PROMOTIONS

Coupons

- Unified engine for sports + cafe: percentage (with cap) or flat; min-amount, validity windows, usage limits per user/total.
- Platform targeting (web-only / app-only / both) and special conditions like FIRST_APP_BOOKING (powers APPFIRST).
- User-group targeting: cohorts defined once under Settings → User Groups feed coupons AND push targeting.
- Public/hidden toggle; hidden codes work but don't show in browse.

Rewards configuration

- Earn rates per sport and for cafe, redeem rate, min redeem, max-% cap per order.
- Signup / referral / birthday bonus amounts; manual point adjustments per user with audit trail.

20. ADMIN — PAYMENTS

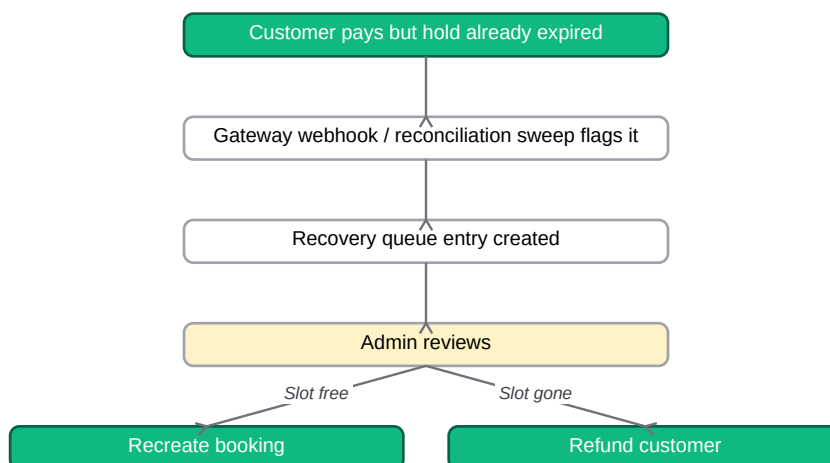
Payment Settings (live switches, no deploy)

- Online payments master toggle · Pay-50%-advance toggle.
- UPI QR mode: **Static QR** or **Dynamic QR (PhonePe)** — mutually exclusive; DQR carries a nested **UPI Intent** toggle (tap-to-pay app list vs scan-only).
- Active gateway selector (Razorpay / PhonePe) drives every checkout on web + app instantly.

Dashboards & recovery

- Razorpay dashboard: live transactions from the gateway API.
- PhonePe dashboard: DQR + static QR transaction views with per-store tabs (Online / Offline / Gym / Yoga / Cafe).
- **Payment Recovery**: webhook + sweep catch payments captured without a booking (hold expired mid-payment); admin resolves to booking or refund — no customer money can silently vanish.

FLOW: Orphaned payment recovery



21. ADMIN — EXPENSES & RUNNING EXPENSES

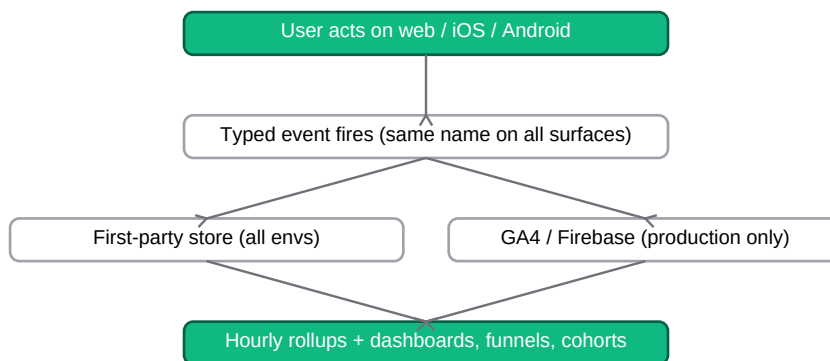
- **Expenses (legacy)**: historical ledger, now read-only — no new entries or edits.
- **Running Expenses**: active ledger with admin-defined dropdown configs (category, payment mode...), month-wise collapsible table, add/edit via modal (desktop) / bottom-sheet (mobile), filters.

- Separate analytics for each module: totals, category breakdowns, trends.
- Available on web admin and the admin app.

22. ADMIN — ANALYTICS & INSIGHTS

- **Business analytics:** revenue/bookings KPIs, sport-wise & cafe breakdowns, peak-hours, payment-method mix, top customers, date-range/grouping filters.
- **Demand insight:** waitlist pressure and slot-unavailable taps show where capacity is short.
- **Funnels:** booking funnel (sport → slots → checkout → paid) with drop-off per step.
- **Cohorts:** user cohorts frozen at first booking; retention views.
- **Events explorer:** every first-party event (web + app) searchable with properties, sessions and user attribution.
- **Push analytics:** delivery/open stats per campaign.
- **Reports:** scheduled/queued report generation (processed by cron).
- **Google Analytics 4** runs in parallel on web + iOS + Android (Firebase), with identical event names across surfaces.

FLOW: Analytics data pipeline



23. MOBILE APP PLATFORM

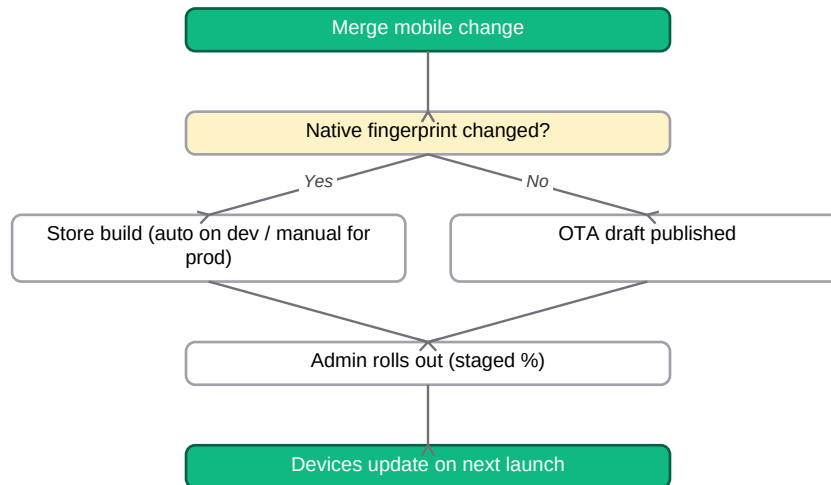
Self-hosted OTA updates

- JS-level app updates ship over-the-air from our own server — no store review for UI/logic changes.
- Admin OTA dashboard: draft releases per channel, staged percentage rollouts, promote/rollback, device adoption stats.
- Updates are code-signed; apps auto-apply pending updates on launch.

Release pipeline

- Every push with mobile changes auto-publishes an OTA draft; a native fingerprint decides JS-only vs native.
- Native changes on the development branch auto-trigger store builds (TestFlight / Play internal); production store builds are a deliberate manual dispatch.
- Release Flow page documents the whole pipeline; version gates let admin force or suggest app updates.

FLOW: Ship an app change



24. INVOICES & DOCUMENTS

- Booking invoice PDF on company letterhead with GST breakdown (download from booking detail, web).
- Cafe order invoice PDF in the same format.
- Public cafe menu PDF endpoint for QR-code table cards.
- Add-to-calendar (ICS) export for bookings in the app.

— END OF DOCUMENT —